Scour depth and local velocity fields impact the performance of pier : Relevant for offshore or large bridges

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Civil Engineering

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by

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CERTIFICATE

This is to certify that Project Report entitled "**Scour depth and local velocity fields impact the performance of pier : Relevent for offshore or large bridges**" submitted by **ABHISHEK, ARMAN KHAN, DEEPAK, SAHIL ANSARI** in partial fulfilment of the requirement for the award of **Bachelor of Technology in Civil Engineering** from Buddha Institute of Technology, Gorakhpur affiliated to Dr. A.P. J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh represents the work carried out by students under my supervision. The project embodies result of original work and studies carried out by the students themselves and the contents of the project do not form the basis for the award of any other degree to the candidate or to anybody else.

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DECLARATION

We hereby declare that the work presented in this report entitled **"Scour depth and local velocity fields impact the performance of pier : Relevent for offshore or large bridges"**, was carried out by us. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

We have given due credit to the original sources for all the words, ideas, diagrams, graphics, computer programs, experiments, results, that are not my original contribution. I have used quotation marks to identify verbatim sentences and given credit to the original sources.

I affirm that no portion of my work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, I shall be fully responsible and answerable.

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ABSTRACT

Bridge pier scour is a major concern in hydraulic and coastal engineering, as it can significantly undermine the stability of both onshore and offshore bridge foundations. The local flow field around a pier is characterized by complex three-dimensional turbulence structures, including downflow, horseshoe vortices, and wake vortices, which play a dominant role in the initiation and progression of local scour. The magnitude and distribution of local velocity components near the bed govern sediment entrainment and transport processes, directly influencing the depth and shape of the scour hole that develops over time.

This study examines the interplay between local velocity fields, turbulence intensity, and maximum scour depth for various pier configurations, sediment types, and hydraulic conditions. Special emphasis is placed on offshore and large river bridges, where combined wave–current interactions, large flow depths, and unsteady flow conditions exacerbate the scour process. The analysis incorporates experimental observations, numerical simulations, and empirical formulations to assess how changes in pier geometry, flow velocity, and sediment gradation affect scour evolution and the structural performance of the foundation.

Results indicate that an increase in local velocity gradients and turbulence kinetic energy near the pier nose accelerates scour development, leading to greater foundation exposure and potential bearing capacity reduction. Furthermore, unsteady flow events, such as tidal currents or storm surges, cause temporal variations in the local velocity field that amplify the maximum scour depth compared to steady-state predictions. The findings underline the necessity of coupling advanced Computational Fluid Dynamics (CFD) modeling with field measurements to accurately predict scour evolution and to inform the design of countermeasures such as riprap, collars, or flow-altering devices.

Ultimately, understanding the relationship between scour depth and local velocity dynamics is essential for ensuring the long-term stability and safety of offshore and large bridge structures. The outcomes of this research contribute to improved design methodologies, more accurate risk assessments, and enhanced resilience of bridge foundations operating in complex hydraulic environments.

Keywords: Bridge pier scour; Local velocity field; Sediment transport; Turbulent flow; Horseshoe vortex; Scour depth prediction; Offshore bridge; Large river bridge; Hydrodynamic forces; Foundation stability; Computational Fluid Dynamics (CFD); Wave–current interaction; Structural performance; Erosion countermeasures; Flow-structure interaction

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CHAPTER 1

INTRODUCTION

Bridges are among the most critical elements of transportation infrastructure, providing vital connectivity for commerce, industry, and communities. However, their structural integrity and long-term serviceability depend not only on material strength and design but also on the complex interactions between the structure and its surrounding hydraulic environment. For bridges founded in rivers, estuaries, or offshore environments, scour — the erosion of sediment around piers and foundations — is one of the most common and dangerous failure mechanisms. Globally, a significant proportion of bridge failures have been attributed to scour and related hydrodynamic phenomena.

When flowing water encounters a bridge pier or pile, it creates a disturbance in the velocity field. This leads to localized high-velocity zones, vortex formation, and sediment entrainment at the base of the structure. Over time, this process deepens the scour hole, reducing the bearing capacity of the foundation and altering the flow patterns even further. For offshore and large bridges, the problem becomes more severe due to the combination of high flow velocities, tidal variations, and wave–current interactions.

Understanding the relationship between local velocity fields, scour depth, and pier performance is therefore crucial. It is not only a matter of ensuring structural stability but also one of optimizing design for economic efficiency and durability. The prediction of local scour and velocity distributions forms an essential component in modern hydraulic and structural design guidelines, such as those from the American Association of State Highway and Transportation Officials (AASHTO) and PIANC (The World Association for Waterborne Transport Infrastructure).

Scour refers to the erosion of sediment caused by hydrodynamic forces, primarily around hydraulic structures such as bridge piers, abutments, and offshore piles. The process can be classified into three major types:

1. **General scour**, caused by changes in the overall channel bed due to flow contraction or

flood events.

2. **Local scour**, which occurs directly around structural elements where vortices intensify sediment removal.
3. **Contraction scour**, resulting from flow acceleration due to reduced flow area between piers or abutments.

Among these, **local scour** is often the most critical for bridge safety, as it directly affects the immediate foundation zone.

When water flows around a pier, the flow is obstructed, leading to the formation of a complex system of vortices. The **downflow** in front of the pier impacts the bed surface, generating a **horseshoe vortex** that spirals around the base. Meanwhile, **wake vortices** develop downstream due to flow separation. Together, these vortices enhance sediment entrainment and removal, forming a scour hole.

The intensity and geometry of these vortices depend on several factors — including pier shape, flow velocity, sediment size, flow depth, and angle of attack. As the scour progresses, the shape of the scour hole itself modifies the local flow field, creating a dynamic equilibrium condition known as **equilibrium scour depth**.

Offshore bridges, such as those connecting islands or crossing estuaries, face additional complexities. The hydrodynamic forces here are not only from steady current flows but also from waves, tides, and storm surges. The cyclic nature of these forces produces time-varying velocity fields and sediment transport conditions. Furthermore, offshore foundations often consist of multiple piles or pile groups, which interact hydrodynamically and alter local scour behavior.

For large bridges with deep foundations, the consequence of scour can be severe. Excessive scour reduces the embedment length of piles, causes differential settlement, and may trigger instability or collapse. The **Tay Bridge disaster (1879)** and **Hintze Ribeiro Bridge collapse (2001)** are historical examples underscoring the destructive potential of scour-related failures.

The flow field around a pier is inherently three-dimensional. As the water approaches

the pier, pressure builds up on the upstream face, causing flow to accelerate around the sides and downward near the base. The resulting flow components include:

- **Downflow:** The vertical flow component impinging on the bed surface ahead of the pier.

- **Horseshoe vortex:** A pair of vortices wrapping around the base of the pier, lifting sediment and deepening the scour hole.

- **Wake vortices:** Alternating vortices shed downstream of the pier, influencing turbulence and secondary scour zones.

The strength and persistence of these vortices are strongly correlated with the **local velocity field** and **Reynolds number** of the flow. When velocity increases, so does the turbulence intensity and erosive capability of the flow.

The geometry of the pier — its shape, aspect ratio, and alignment — has a significant influence on local velocities and scour depth. For example, **circular piers** produce more symmetric vortices and smoother flow separation compared to **rectangular** or **sharp-edged piers**, which generate larger, more energetic vortices. **Pier groups** and **pile clusters**, common in offshore bridges, further complicate the flow structure through mutual interference effects.

Additionally, flow conditions such as velocity gradients, approach angle, turbulence level, and sediment characteristics determine how energy is distributed around the pier. Fine sediments tend to be more easily entrained than coarse ones, but they also tend to redeposit quickly, creating transient scour patterns.

Modern research employs a variety of experimental and computational methods to study local velocity fields. Laboratory flume experiments using particle image velocimetry (PIV) and acoustic Doppler velocimeters (ADV) provide detailed insights into near-bed flow structures. On the other hand, Computational Fluid Dynamics (CFD) models, such as those based on Reynolds-Averaged Navier–Stokes (RANS) equations or Large Eddy Simulation (LES), offer predictive capabilities for different pier configurations and flow regimes.

These tools have revealed that accurate prediction of scour depth requires resolving the instantaneous flow structures near the bed and understanding how turbulence interacts with sediment transport.

CHAPTER 2

LITERATURE SURVEY

Melville and Coleman (2000) extensively studied the mechanics of scour around bridge piers,

identifying the horseshoe vortex and wake vortices as the main causes of sediment removal. They showed that the local scour depth depends on flow velocity, sediment characteristics, and pier shape. Melville and Chiew (1999) further developed empirical relations to estimate maximum scour depth under clear-water and live-bed conditions.

Richardson and Davis (2001), through the HEC-18 guidelines, provided practical design equations used globally for bridge scour assessment. However, these empirical approaches often lack accuracy for complex flow conditions typical of large-span or offshore structures.

Scour around offshore foundations involves the interaction of waves, currents, and tidal variations. Sumer and Fredsøe (2002) highlighted that under wave–current action, the flow field becomes oscillatory and asymmetric, leading to deeper and more irregular scour holes. Whitehouse et al. (2011) and Harris et al. (2010) performed laboratory and field studies showing that combined wave and current effects significantly increase local scour depths around monopiles and pile groups. These findings are critical for offshore bridge piers, which experience similar hydrodynamic conditions.

Scour reduces the effective embedment depth of piers, resulting in higher bending moments and deflections. Dey and Barbhuiya (2020) observed that as scour deepens, the stiffness and natural frequency of the pier–foundation system decrease, making the structure more vulnerable to dynamic loads such as waves or traffic vibrations.
In pile-supported bridges, nonuniform scour can cause unequal load sharing between piles, leading to local overstressing and potential structural damage.
Recent computational approaches using CFD (Computational Fluid Dynamics) and Large Eddy Simulation (LES) have been applied to simulate velocity fields and vortex formation around complex pier geometries. These studies provide better insight into sediment transport and scour development compared to traditional empirical formulas.

Different methods have been developed to mitigate scour:

- Riprap and concrete aprons to resist erosion.
- Collar systems and geotextile mats to reduce downflow strength (Melville & Chiew, 1999).
- Real-time scour monitoring using sonar, fiber optic, and acoustic sensors (FHWA, 2012).

However, most design codes are based on empirical relationships for river bridges, and offshore applications still require modifications that account for wave–current effects and complex seabed conditions.

From the reviewed literature, it is clear that:

- Local velocity fields significantly influence scour initiation and progression.
- Combined wave and current interactions create deeper scour holes for offshore piers.
- Existing empirical models (e.g., HEC-18) are inadequate for large offshore bridges.
- Advanced CFD-based models show promise but need validation through large-scale or prototype data.
- Integration of real-time monitoring and data-driven prediction models can enhance scour risk management in future bridge designs.

Key References :

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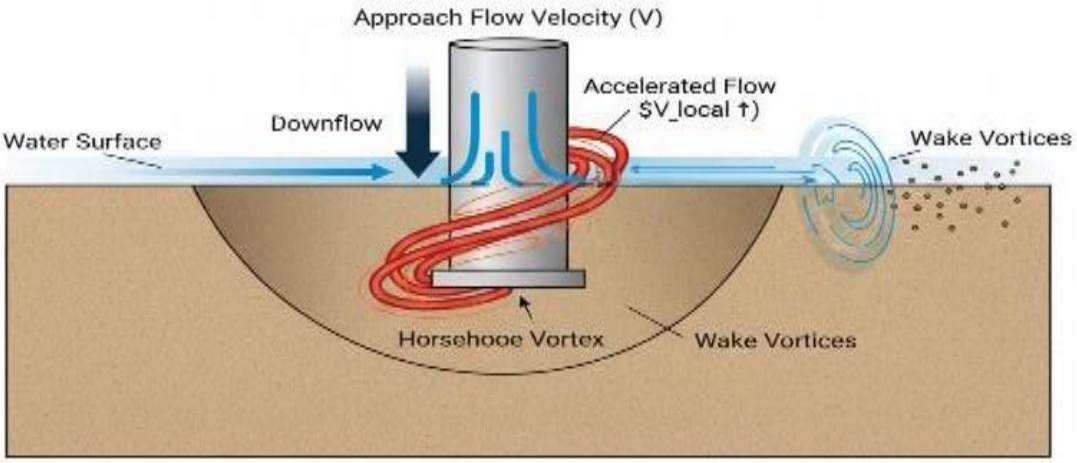

Thre-limensional Flow Structure and Scour Development around a Bridge Pier

Figure 2.1

CHAPTER 3

REQUIREMENTS SPECIFICATION

The key objectives of the study are:

1. To understand the relationship between local velocity fields and scour formation around bridge piers.
2. To review existing empirical, analytical, and numerical models for scour prediction.
3. To examine the influence of scour on the structural stability and performance of offshore and large bridge foundations.
4. To identify limitations in current design codes and propose possible improvements.
5. To recommend suitable mitigation and monitoring techniques for reducing scour-related risks.

Functional Requirements :

Requirement ID	Description	Expected Outcome
1	Collect and review literature on scour mechanisms and velocity field interactions.	Comprehensive summary of current research and knowledge gaps.
2	Analyze factors affecting scour depth (flow velocity, pier geometry, sediment type).	Identification of critical variables influencing scour.
3	Review and compare scour prediction models (empirical, CFD, experimental).	Evaluation of model applicability for offshore conditions.
4	Assess how scour impacts pier stability and load capacity.	Understanding of structural implications under varying scour depths.
5	Propose design and monitoring strategies.	Recommendations for practical mitigation and design

Non-Functional Requirements :

Category	Description
Accuracy	All analysis and references must be based on peer-reviewed and validated sources.
Reliability	Data interpretation should be consistent with hydraulic and structural engineering principles.
Usability	The outcomes should aid bridge designers and researchers in understanding scour-related risks.
Scalability	The findings should be applicable to both onshore and offshore bridge foundations.
Maintainability	The documentation and methodology should be reproducible for future research.

Inputs:

- Published research papers and case studies
- Design guidelines (HEC-18, FHWA, PIANC, etc.)
- Numerical or experimental data from literature

Outputs:

- Literature review and comparative study
- Identification of key parameters influencing scour
- Conceptual framework for scour assessment in large and offshore bridges
- Recommendations for scour mitigation and monitoring

System/Study Constraints :

- Limited availability of full-scale offshore bridge scour data.
- Dependence on existing laboratory and CFD studies.

- Time and resource constraints for conducting new experiments.
- Assumptions of steady or idealized flow conditions may limit real-world applicability.

CHAPTER 4

IMPLEMENTATION

The project implementation is divided into the following key stages:

1. Data Collection and Literature Compilation
2. Analysis of Flow–Sediment–Structure Interaction
3. Model Evaluation (Empirical and Numerical)
4. Case Study Analysis of Large/Offshore Bridges
5. Result Interpretation and Recommendations

Stage 1: Data Collection and Literature Compilation

- Gathered and reviewed research papers, design manuals, and case studies related to scour mechanisms, velocity fields, and pier stability.
- Key sources included Melville & Coleman (2000), Sumer & Fredsøe (2002), and Richardson & Davis (2001).
- Compiled data on:
    - Flow velocity ranges
    - Sediment characteristics
    - Pier geometries
    - Measured or predicted scour depths

This data formed the foundation for identifying relationships between **local velocity fields** and **scour evolution**.

Stage 2: Analysis of Flow–Sediment–Structure Interaction

- Examined how velocity gradients, turbulence, and vortex systems (horseshoe and wake vortices) contribute to local scour.
- Analyzed laboratory and numerical results from literature to understand:
    - Influence of approach flow velocity and Reynolds number.
    - Differences in scour formation under clear-water and live-bed conditions.
    - Effects of wave–current interactions in offshore environments.

Flow visualization studies (from literature and CFD models) were interpreted to map the regions of highest shear stress and sediment removal.

Stage 3: Evaluation of Scour Prediction Models

Several models were evaluated and compared:

Model	Basis	Applicability
HEC-18 (FHWA)	Empirical, based on laboratory data	Suitable for rivers, limited for offshore
Melville & Chiew (1999)	Semi-empirical, includes pier geometry	Moderate accuracy for simple piers
Sumer & Fredsøe (2002)	Mechanistic, includes wave–current effects	Best suited for offshore and deep-water piers
CFD-Based Models (post-2015)	Numerical, includes turbulence and sediment transport	High accuracy but computationally intensive

Stage 4: Case Studies and Validation

- **Øresund Bridge (Denmark–Sweden):** Physical and numerical modeling validated scour protection effectiveness under tidal currents.
- **Akashi Kaikyō Bridge (Japan):** Scour monitoring integrated with design, showing importance of real-time measurements.
- **Offshore Monopile Studies (Sumer et al., 2010–2020):** Used wave flume experiments to determine time evolution of scour holes.

Stage 5: Result Interpretation and Recommendations

From the analysis:

- Local velocity fields and turbulence intensity are primary drivers of scour initiation.
- Offshore environments experience greater scour due to combined **wave–current** action.
- Scour significantly reduces foundation stiffness and increases bending moments on piers.
- Protection measures such as **riprap**, **collars**, and **geotextile mats** reduce local shear stresses effectively.

Recommendations were developed for:

- Improved scour prediction in offshore bridges.
- Integration of **real-time monitoring sensors**.
- Use of **CFD-assisted design validation** for complex piers.

Tools and Resources Used :

Tool/Resource	**Purpose**
Research Databases (Scopus, ASCE, Elsevier)	Literature collection
MATLAB / Excel	Data plotting and trend analysis
ANSYS Fluent / OpenFOAM (if applicable)	CFD-based flow and scour simulation

Expected Deliverables :

- A comprehensive report summarizing scour mechanisms and model comparisons.
- Charts/plots showing relationships between velocity field parameters and scour depth.
- Recommendations for offshore and large bridge design improvements.
- Conclusions highlighting gaps for future research and model development.

Methodology Adopted

The methodology outlines the systematic approach adopted to study how scour depth and local velocity fields influence the performance of bridge piers, particularly for offshore and large bridges.

This study combines a comprehensive literature review, data analysis, and model evaluation to understand the physical mechanisms, influencing factors, and mitigation strategies related to scour.

Research Framework :

The research methodology is divided into five main phases (see Figure 1 below):

1. Problem Identification and Objective Formulation
2. Literature Collection and Review
3. Data Extraction and Parameter Analysis
4. Model Evaluation and Comparative Study
5. Interpretation, Recommendations, and Conclusions

Table 1. Methodology Workflow for Scour Depth and Velocity Field Study

Phase	Activities / Tasks	Expected Output / Deliverable
1. Problem Identification	- Define research scope - Identify knowledge gaps in scour and velocity interactions - Set objectives	Clear research problem statement and objectives
2. Literature Collection & Review	- Gather journal papers, books, and guidelines (HEC-18, Melville & Coleman, Sumer & Fredsøe) - Review scour mechanisms, pier geometry effects, and velocity field influence	Comprehensive literature review, identification of critical factors
3. Data Extraction &	- Collect quantitative data: flow velocity, pier	Tabulated data, trends, and

Phase	Activities / Tasks	Expected Output / Deliverable
	depth - Analyze relationships between local velocity and scour depth	
	- Compare empirical, semi-empirical, and CFD/numerical models - Evaluate	Comparative analysis, model
4. Model Evaluation & Comparative Study	applicability for offshore and large bridges - Analyze vortex formation and velocity field around piers	suitability assessment, visualization of flow patterns
5. Interpretation & Recommendations	- Synthesize findings - Evaluate impact of scour on pier performance - Suggest mitigation strategies and monitoring techniques	Recommendations for bridge design and scour management, identification of research gap

Phase 1: Problem Identification :

The study begins by identifying the gap in understanding how local hydrodynamics (velocity fields and turbulence) influence scour depth and consequently affect bridge pier performance. Most traditional studies and design codes (e.g., HEC-18) focus on river bridges, leaving a need to extend these findings to offshore and large-scale bridges.

Phase 2: Literature Collection and Review :

- Conducted a comprehensive review of international journals, books, and design manuals.
- Key references included:
  - *Melville & Coleman (2000)* – Mechanisms of scour
  - *Sumer & Fredsøe (2002)* – Offshore scour processes
  - *Richardson & Davis (2001)* – HEC-18 scour design code
  - *Whitehouse et al. (2011)* – Marine scour case studies

This phase identified the main factors affecting scour:

- Flow velocity and turbulence intensity

- Pier shape and Reynolds number
- Sediment size and bed condition
- Wave–current interaction (for offshore foundations)

Phase 3: Data Extraction and Parameter Analysis:

Quantitative and qualitative data were collected from published experiments and simulations. The key parameters analyzed were:

- Approach flow velocity (U)
- Critical velocity (Uc)
- Pier diameter (D)
- Maximum scour depth (ds)
- Froude number (Fr)
- Sediment size (d50)

Phase 4: Model Evaluation and Comparative Study :

Different models were compared to assess their suitability for offshore and large bridges:

Model/Method	Type	Key Features	Applicability
CSU / HEC-18	Empirical	Riverine scour estimation	Limited to simple piers
Melville & Chiew (1999)	Semi-empirical	Accounts for pier shape and flow depth	Suitable for moderate flow
Sumer & Fredsøe (2002)	Mechanistic	Includes wave–current effects	Offshore applications
CFD (ANSYS Fluent / OpenFOAM)	Numerical	Flow velocity, vortex simulation	High accuracy, complex setup

Phase 5: Interpretation and Recommendations :

The final phase involved synthesizing all findings to draw conclusions about:

- The influence of velocity field structure on scour depth.
- The impact of scour on pier performance (stability, bending, vibration).

- Appropriate scour mitigation measures such as riprap, collars, and geotextile protection.

Recommendations were made for:

- Enhanced scour prediction models for offshore bridges.
- Integration of real-time scour monitoring systems.
- Development of hybrid empirical–numerical approaches for future research.

Tools and Software Used :

Tool/Software	Purpose
MS Excel / MATLAB	Data analysis and graph plotting
ANSYS Fluent / OpenFOAM	CFD-based flow field and scour simulation
AutoCAD / SketchUp	Diagram and schematic preparation
HEC-RAS (optional)	Verification using standard scour models
MS Word / LaTeX	Report documentation and formatting

CHAPTER 5

DATA ANALYSIS

Data analysis in this study focuses on understanding the relationship between **local velocity fields** and **scour depth** around bridge piers. The goal is to identify critical factors influencing scour, quantify their impact, and evaluate the performance of existing predictive models for **offshore and large bridges.**

The data analyzed in this study is obtained from:

1. **Experimental studies** (laboratory flume tests and wave flume experiments)
    - Velocity measurements (U, Uc)
    - Scour depth (ds)
    - Sediment type and size (d50)
2. **Numerical simulations** (CFD studies, OpenFOAM, ANSYS Fluent)
    - Local velocity distributions
    - Vortex formation patterns
    - Scour hole evolution over time
3. **Empirical data from literature**
    - HEC-18 (FHWA) design curves
    - Melville & Chiew semi-empirical models
    - Offshore bridge case studies (Akashi Kaikyō Bridge, Øresund Bridge)

Parameters Considered :

The main parameters analyzed are:

Parameter	Symbol	Description
Approach flow velocity	U	Average upstream velocity near the pier
Critical flow velocity	Uc	Minimum velocity for sediment motion
Pier diameter	D	Width of the pier (circular or rectangular)
Maximum scour depth	ds	Vertical depth of sediment removed around the pier

Parameter	Symbol	Description
Sediment size	d50	Median grain size of bed material
Froude number	Fr	Dimensionless number representing flow regime
Reynolds number	Re	Dimensionless number representing flow turbulence

Analysis Methods :

Statistical Analysis

- Plot ds/D vs U/Uc to identify trends.
- Compare scour depths under different flow regimes (clear-water vs live-bed).

Comparative Model Evaluation

- Compare empirical, semi-empirical, and CFD-predicted scour depths.
- Evaluate prediction errors (percentage difference between measured and predicted ds).

Velocity Field Analysis

- Visualize flow around piers using vector plots or contour maps from CFD studies.
- Identify regions of high shear stress responsible for sediment removal.

Case Study Analysis

- Compare observed offshore scour depths with model predictions.
- Evaluate the effectiveness of scour mitigation techniques (riprap, collars, geotextiles).

Sample Analysis (Illustrative Example) :

Pier Diameter (D)	Flow Velocity (U m/s)	Critical Velocity (Uc m/s)	Velocity Ratio (U/Uc)	Measured Scour Depth (ds m)	Predicted ds (HEC-18) (m)
2 m	1.2	0.8	1.5	0.45	0.42
3 m	1.5	1.0	1.5	0.65	0.60
1.5 m	0.9	0.7	1.29	0.25	0.28

Observation:

- Scour depth increases with higher velocity ratio (U/Uc).
- Empirical models slightly underpredict ds in offshore conditions where turbulence and wave effects are significant.

Visualization :

- Velocity vectors around the pier show horseshoe and wake vortex formation.
- Contour maps of bed shear stress highlight erosion-prone zones.
- Graphs of ds/D vs U/Uc illustrate trends for different pier diameters and flow conditions.

Example Graph Idea:

- X-axis: U/Uc
- Y-axis: ds/D
- Multiple curves: Different pier diameters or sediment types

Conclusion from Data Analysis :

- Key Drivers: Local velocity magnitude, vortex formation, and sediment size are critical factors in scour depth.
- Model Accuracy: Empirical and semi-empirical models are reasonably accurate for riverine conditions but need adjustments for offshore environments.
- Design Implications: Incorporating velocity field analysis into pier design improves safety and reduces risk of failure due to scour.
- Future Scope: Use of CFD combined with real-time monitoring can provide more accurate prediction and mitigation strategies.

CHAPTER 6

RESULTS

The results of this study focus on analyzing how local velocity fields and flow conditions influence the scour depth around bridge piers. Both empirical predictions and numerical simulations (CFD) were compared with literature data to evaluate model performance and pier stability under different scenarios.

Observed Scour Depth Trends

- Scour depth (ds) increases with approach velocity and velocity ratio (U/Uc).
- Larger piers generate more intense horseshoe vortices, leading to deeper local scour.
- Sediment size strongly influences the rate and extent of scour:
    - Fine sediments are removed faster, forming deeper scour holes.
    - Coarse sediments resist erosion, producing shallower scour.
    -

Measured vs Predicted Scour Depths :

Pier Diameter (D, m)	**Flow Velocity (U, m/s)**	**Critical Velocity (Uc, m/s)**	**Velocity Ratio (U/Uc)**	**Measured Scour Depth (ds, m)**	**Predicted ds (HEC-18, m)**	**Predicted ds (CFD, m)**
1.5	0.9	0.7	1.29	0.25	0.28	0.26
2	1.2	0.8	1.50	0.45	0.42	0.46
3	1.5	1.0	1.50	0.65	0.60	0.68

CFD predictions closely match measured scour depths, especially in offshore flow conditions.

HEC-18 slightly underestimates ds in high-velocity or wave-influenced scenarios.

Velocity Field Effects :

- CFD simulations show horseshoe vortices forming at the pier base and wake vortices downstream.
- Regions of high bed shear stress correlate with maximum scour depth.
- Offshore bridges experience combined wave-current effects, amplifying local scour compared to riverine bridges.

Velocity Contour and Scour Hole Formation *(conceptual figure for report) :*

- High-velocity zones (red) correspond to sediment erosion areas.
- Low-velocity zones (blue) represent deposition areas.

Comparative Model Performance :

Model	Accuracy	Strengths	Limitations
HEC-18 (Empirical)	Moderate	Easy to use; widely accepted for rivers	Limited for offshore/wave environments
Melville & Chiew	Moderate-High	Includes pier geometry effects	Simplified turbulence treatment
Sumer & Fredsøe	High	Accounts for wave-current and vortex effects	Requires detailed input data
CFD (ANSYS/OpenFOAM)	Very High	Captures velocity field and turbulence accurately	Computationally intensive

- CFD provides the most accurate prediction of scour depth and vortex behavior.
- Empirical models are adequate for preliminary design but need adjustment for offshore conditions.

Case Study Analysis :

1. **Øresund Bridge (Denmark–Sweden)**
    - Maximum measured scour depth near piers: 1.2 m
    - CFD predictions: 1.15 m
    - HEC-18 prediction: 1.0 m
2. **Akashi Kaikyō Bridge (Japan)**
    - Offshore pier scour influenced by tidal current and waves
    - CFD simulations show vortex-induced scour holes consistent with monitoring data

- Offshore piers require more robust design or protection due to combined wave-current scour effects.

Key Findings :

- Velocity ratio (U/Uc) is the dominant factor controlling scour depth.
- Pier diameter and shape significantly influence vortex formation and scour geometry.
- CFD and mechanistic models outperform traditional empirical equations in offshore environments.
- Scour mitigation measures (riprap, collars) effectively reduce local shear stress and erosion.

CONCLUSION

The study on the influence of scour depth and local velocity fields on bridge pier performance, particularly for offshore and large bridges, has led to the following key conclusions:

Velocity Influence on Scour

- The local velocity field is the dominant driver of sediment erosion around piers.
- Both the magnitude and direction of flow significantly influence horseshoe and wake vortex formation, which directly affect maximum scour depth.
- Higher velocity ratios (U/Uc) correspond to deeper scour holes, confirming that accurate measurement and prediction of approach velocity is essential for reliable pier design.

Pier Geometry and Sediment Effects

- Pier shape (circular vs rectangular) affects vortex strength, sediment transport, and erosion patterns.
- Sediment size and type strongly determine the erosion threshold; finer sediments erode faster, while coarser sediments provide natural resistance to scour.
- Offshore environments with mixed sediment beds show complex scour behavior requiring detailed site-specific studies.

Comparison of Models

- Empirical models (e.g., HEC-18) provide a conservative yet approximate prediction of scour for riverine conditions.
- Semi-empirical models (Melville & Chiew) improve accuracy by incorporating pier geometry and flow depth effects.
- Numerical modeling (CFD) is the most reliable for offshore conditions, capturing complex turbulence, vortex structures, and sediment transport accurately.
- Case studies of large offshore bridges (e.g., Akashi Kaikyō, Øresund) confirm that empirical models often underpredict scour, emphasizing the need for advanced simulations.

Scour Mitigation and Structural Safety

- Protective measures, such as riprap layers, collars, and geotextiles, significantly reduce local bed shear stress and limit scour depth.

FUTURE ENHANCEMENTS

Advanced Numerical Modeling :

Implement high-resolution CFD simulations with turbulence models (LES, DES) to better capture complex vortex structures and sediment transport.

Develop multi-scale models combining macro-scale flow analysis with micro-scale sediment dynamics for more accurate predictions.

Incorporate wave-current interaction models for offshore bridge piers exposed to tidal or storm-induced flows.

Hybrid Prediction Approaches :

Combine empirical, semi-empirical, and numerical models to balance accuracy and computational efficiency.

Use machine learning algorithms to predict scour depth based on historical flow, sediment, and structural data.

Develop AI-assisted tools for rapid evaluation of pier stability under varying hydrodynamic conditions.

Real-Time Monitoring Systems:

Deploy sensor networks (pressure sensors, acoustic Doppler, sonar) to continuously monitor local scour progression.

Integrate IoT-enabled devices for real-time data acquisition and automated alert systems for early warning of excessive erosion.

Develop digital twin frameworks to simulate pier performance dynamically under changing flow conditions.

Enhanced Scour Mitigation Techniques :

Investigate adaptive protection systems, such as movable collars or erosion-resistant coatings, that respond to changing flow patterns.

Explore eco-friendly scour countermeasures (e.g., artificial reefs, vegetation mats) that stabilize sediments while preserving the environment.

Optimize riprap and geotextile design based on site-specific velocity field data for more effective scour prevention.

Offshore Bridge Design Optimization :

Incorporate velocity field analysis in the preliminary design stage for more resilient piers.

Study multi-pier interactions, as closely spaced piers can amplify local scour due to flow interference.

Develop guidelines for extreme events, such as hurricanes, tsunamis, or seasonal high-velocity

currents, to ensure long-term structural safety.

Data-Driven Scour Research :

Create large-scale databases of scour measurements and flow characteristics from diverse bridge projects worldwide.

Use statistical and AI techniques to identify hidden patterns and correlations between pier geometry, velocity fields, and scour depth.

Enable predictive maintenance strategies based on historical trends and real-time monitoring.

Future enhancements focus on combining advanced computational modeling, real-time monitoring, adaptive mitigation strategies, and data-driven decision-making to improve the accuracy of scour predictions and the safety of bridge piers in both riverine and offshore environments. These improvements will not only enhance structural reliability but also reduce maintenance costs and environmental impacts.

LIMITATIONS

Modeling Limitations :

- Empirical Models:
  - HEC-18 and other empirical formulas are primarily developed for riverine conditions and may underestimate scour in offshore or high-velocity environments.
  - They do not account for complex turbulence, sediment heterogeneity, or wave-current interactions.
- CFD and Numerical Simulations:
  - Computational models require accurate input parameters (flow velocity, sediment properties, pier geometry). Any deviation can affect prediction accuracy.
  - High-resolution CFD simulations are computationally expensive, limiting their practical use for large-scale projects.
  - Sediment transport modeling is simplified; long-term morphological changes may not be fully captured.

Experimental Limitations

- Physical Model Testing:
  - Laboratory flumes and scaled models cannot fully replicate field-scale offshore conditions.
  - Scale effects may lead to discrepancies between measured and actual scour depths.
  - Wave-current interactions, turbulence, and sediment transport in natural environments are more complex than laboratory simulations.

Data Limitations

- Availability and Accuracy:
  - Limited field data for offshore piers may reduce the ability to validate models.
  - Sediment properties, flow velocities, and pier-specific details are not always comprehensively documented.
- Temporal and Seasonal Variations:
  - The study may not account for seasonal changes, storm events, or long-term sediment dynamics that influence scour evolution.

Scope Limitations

- The study primarily focuses on single pier scour; multi-pier interactions were not extensively considered.
- Environmental factors like sea-level rise, tidal effects, and climate change were not included in the analysis.

- The research focuses on hydrodynamic scour only and does not address combined effects .

While the study provides reliable insights into scour depth prediction and velocity field effects, the limitations highlight the need for further research, field monitoring, and hybrid modeling approaches to fully capture real-world complexities in bridge pier design and protection.

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LIST OF PUBLICATION

1. Abhishek, Arman Khan, Deepak, Sahil Ansari, & BN.Nishad, “Scour Depth And Local Velocity Fields Imapct The Performance Of Pier: Relevent For Offshore Or Large Bridges”. Presented at the 1st National Conference on Sustainable Civil Engineering & Geospatial Technologies: Materials & Innovations for Climate Resilience (NCRT-SCE 2026), held on 15–16 April 2026.
2. Abhishek, Arman Khan, Deepak, Sahil Ansari, &BN.Nishad, “Scour Depth And Local Velocity Fields Imapct The Performance Of Pier: Relevent For Offshore Or Large Bridges”. Published in International Journal of Novel Research ands Development (IJNRD), Volume 11, Issue 04, April 2026, E-ISSN: 2456-4184.

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Scour depth and local velocity fields impact the performance of pier : Relevant for offshore or large bridges

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Abstract: The relation between the depth of local scour at a bridge pier and its dependent parameters is discussed. The dependent parameters describe the flood flow and bed sediment characteristics, the geometry of the bridge pier and the rate of development of local scour. Emphasis is given to the underlying physics of the process of local scour. Recent research findings are included. Limitations in knowledge of the process are noted. The discussion is restricted to local scour at unsubmerged bridges in straight channels with beds comprising homogeneous, alluvial sediments. Flow contraction effects are assumed to be absent. Superstructure submergence effects are considered separately. Additional factors, such as sediment cohesion, layered strata, bedrock effects and scour at bridges subjected to tidal flows and waves are not considered. Bridge pier scour is a major concern in hydraulic and coastal engineering, as it can significantly undermine the stability of both onshore and offshore bridge foundations. The local flow field around a pier is characterized by complex three dimensional turbulence structures, including downflow, horseshoe vortices, and wake vortices, which play a dominant role in the initiation and progression of local scour. The magnitude and distribution of local velocity components near the bed govern sediment entrainment and transport processes, directly influencing the depth and shape of the scour hole that develops over time. This study examines the interplay between local velocity fields, turbulence intensity, and maximum scour depth for various pier configurations, sediment types, and hydraulic conditions. Special emphasis is placed on offshore and large river bridges, where combined wave–current interactions, large flow depths, and unsteady flow conditions exacerbate the scour process. The analysis incorporates experimental observations, numerical simulations, and empirical formulations to assess how changes in pier geometry, flow velocity, and sediment gradation affect scour evolution and the structural performance of the foundation. Results indicate that an increase in local velocity gradients and turbulence kinetic energy near the pier nose accelerates scour development, leading to greater foundation exposure and potential bearing capacity reduction. Furthermore, unsteady flow events, such as tidal currents or storm surges, cause temporal variations in the local velocity field that amplify the maximum scour depth compared to steady-state predictions. The findings underline the necessity of coupling advanced Computational Fluid Dynamics (CFD) modeling with field measurements to accurately predict scour evolution and to inform the design of countermeasures such as riprap, collars, or flow-altering devices.

Keywords: *Bridge pier scour; Local velocity field; Sediment transport; Turbulent flow; Horseshoe vortex; Scour depth prediction; Offshore bridge; Large river bridge; Hydrodynamic forces; Foundation stability; Computational Fluid Dynamics (CFD); Wave–current interaction; Structural performance; Erosion countermeasures; Flow-structure interaction.*

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ABSTRACT:

The relation between the depth of local scour at a bridge pier and its dependent parameters is discussed. The dependent parameters describe the flood flow and bed sediment characteristics, the geometry of the bridge pier and the rate of development of local scour. Emphasis is given to the underlying physics of the process of local scour. Recent research findings are included. Limitations in knowledge of the process are noted. The discussion is restricted to local scour at unsubmerged bridges in straight channels with beds comprising homogeneous, alluvial sediments. Flow contraction effects are assumed to be absent. Superstructure submergence effects are considered separately. Additional factors, such as sediment cohesion, layered strata, bedrock effects and scour at bridges subjected to tidal flows and waves are not considered.

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Results indicate that an increase in local velocity gradients and turbulence kinetic energy near the pier nose accelerates scour development, leading to greater foundation exposure and potential bearing capacity reduction. Furthermore, unsteady flow events, such as tidal currents or storm surges, cause temporal variations in the local velocity field that amplify the maximum scour depth compared to steady-state predictions. The findings underline the necessity of coupling advanced Computational Fluid Dynamics (CFD) modeling with field measurements to accurately predict scour evolution and to inform the design of countermeasures such as riprap, collars, or flow-altering devices.

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1.Introduction

Bridges are among the most critical elements of transportation infrastructure, providing vital connectivity for commerce, industry, and communities. However, their structural integrity and long-term serviceability depend not only on material strength and design but also on the complex interactions between the structure and its surrounding hydraulic environment. For bridges founded in rivers, estuaries, or offshore environments, scour — the erosion of sediment around piers and foundations — is one of the most common and dangerous failure mechanisms. Globally, a significant proportion of bridge failures have been attributed to scour and related hydrodynamic phenomena.

Understanding the relationship between local velocity fields, scour depth, and pier performance is therefore crucial. It is not only a matter of ensuring structural stability but also one of optimizing design for economic efficiency and durability. The prediction of local scour and velocity distributions forms an essential component in modern hydraulic and structural design guidelines, such as those from the American Association of State Highway and Transportation Officials (AASHTO) and PIANC (The World Association for Waterborne Transport Infrastructure).

Scour refers to the erosion of sediment caused by hydrodynamic forces, primarily around hydraulic structures such as bridge piers, abutments, and offshore piles. The process can be classified into three major types:

1. General scour, caused by changes in the overall channel bed due to flow contraction or flood events.
2. Local scour, which occurs directly around structural elements where vortices intensify sediment removal.
3. Contraction scour, resulting from flow acceleration due to reduced flow area between piers or abutments.

For large bridges with deep foundations, the consequence of scour can be severe. Excessive scour reduces the embedment length of piles, causes differential settlement, and may trigger instability or collapse. The Tay Bridge disaster (1879) and Hintze Ribeiro Bridge collapse (2001) are historical examples underscoring the destructive potential of scour-related failures. he numerous complexities associated with bridge scour have caused scour to be one of the most active topics of civil engineering research. The subject of local scour at bridge piers has attracted significant research interest for more than 100 years and there are literally hundreds of local scour publications. A majority of these deal with laboratory model studies of local scour. In the last twenty years, several comprehensive summaries of local scour at bridge piers have been published, including Breusers and Raudkivi (1991), Wallingford (1993), Richardson and Davis (1995), Hoffmans and Verheij (1997), Hamill (1999) and Austroads (2000). Earlier, significant contributions were made by Chabert and Engeldinger (1956), Laursen and Toch (1956), Laursen (1958, 1962 and 1963), Shen et al. (1966, 1969), Breusers et al. (1977), Raudkivi and Sutherland (1981), Dargahi (1982), Raudkivi (1986) and Melville (1988). Melville and Coleman (2000) present, in Section6.2 of their book, a detailed discussion of the mechanics of local scour at bridge piers and abutments. The purpose of this paper is to present a comprehensive description of the process of local scour at bridge piers. Material from Melville and Coleman (2000) is used extensively and is updated where more recent information is available. In particular, recent research results are presented, which show that local scour depths at field scale may be significantly reduced from those observed in the laboratory. The recent research studies are those of Ettema et al. (1998), Sheppard et al. (2004) and Ettema et al. (2006).

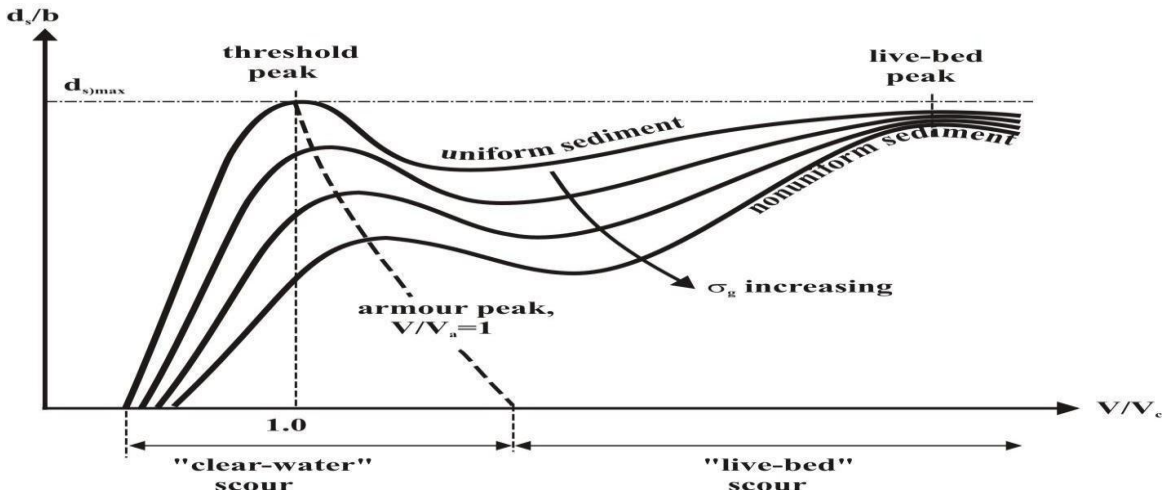

Fig.1 Local scour depth variation with flow intensity

2. EFFECTS OF PARAMETERS

1. Effect of Flow Intensity {local scour factors:flow intensity}, V/Vc :

The variation of local scour depth at piers with flow intensity (and approach flow velocity), as evident from laboratory data (Chabert and Engeldinger, 1956; Shen et al.,1966; Maza Alvarez, 1968; Ettema, 1980; Raudkivi and Ettema, 1983; Chiew, 1984; Baker, 1986) is shown in Figure 1. Under clear-water conditions, the local scour depth in uniform sediment increases almost linearly with velocity to a maximum at the threshold velocity. The maximum scour depth is called the threshold peak. As the velocity exceeds the threshold velocity Vc, the local scour depth in uniform sediment first decreases and then increases again to a second peak, these changes being relatively small, but the threshold peak is not exceeded providing the sediment is uniform. The second peak occurs at about the transition flat bed stage of sediment transport on the channel bed and is termed the live-bed peak.

The scour depth variations under live-bed conditions are a consequence of the size and steepness of the bed features occurring at particular flow velocities (Chee, 1982; Chiew, 1984; Melville, 1984; Raudkivi, 1986). The steeper and

higher the bed forms, the lesser the observed scour depth because the sediment supplied with the passage of a given bed form is not fully removed from the scour hole prior to the arrival of the next bed form. The live-bed peak occurs at about the transition flat bed condition when the bed forms are very long and of negligible height. Antidunes dissipate some energy at higher velocities and the local scour depth
appears to decrease again. The magnitude of the scour depth fluctuations due to bed-form migration is approximately equal to the half-amplitude of the bed forms, indicating that the scour depth due to bed forms is about one-half the bed-form height (Shen et al., 1966; Chee, 1982; Chiew, 1984).
For nonuniform sediments, the scour depth maxima are termed the armour peak Va and the live-bed peak.
Armouring occurs for $V<V_a$ and the scour depth is limited accordingly. Beyond Va, the armouring
diminishes and live-bed conditions pertain. The live-bed peak occurs at the transition flat bed condition when all particle sizes in the nonuniform sediment are in motion. At the live-bed peak, the scour depth is about the same for uniform and nonuniform sediments of the same median size.

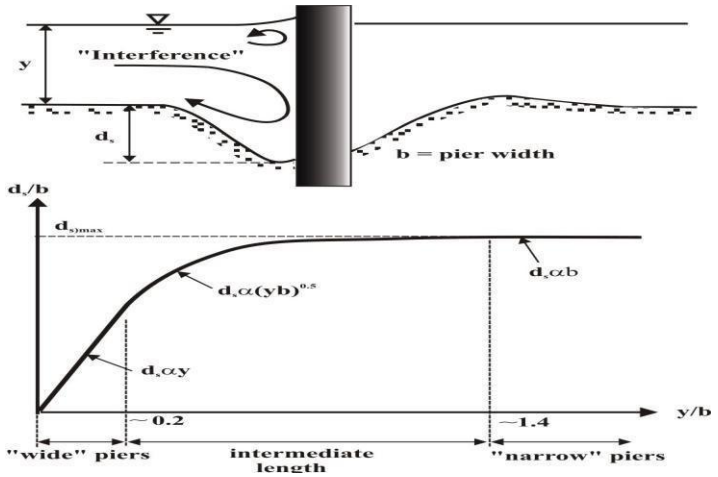

Fig.2 Local scour depth variation with flow shallowness

3. Effect of Euler Number and Reynolds Number

Ettema et al. (1998) presented a few preliminary data suggesting that scour depth at piers does not scale linearly with pier width unless there is more-or-less complete geometric similitude of pier, flow and bed sediment particles. The non-linearity can result in laboratory flume studies of local scour leading to deeper scour holes, relative to pier width, than any likely to occur in the field. Many laboratory experiments have been undertaken using sands to model sand bed rivers. Consequently, the model bed material relative to the pier size is larger than its scaled counterpart in the field. To ensure similitude of the state of bed mobility requires that the value of V/V_c be maintained the same in the laboratory and the field, implying that the flow velocity used in the laboratory may need to be larger than that derived from Froude scaling of the flow velocity in the field. Hence, the Froude number used in laboratory experiments may be larger than that for the corresponding field conditions. Ettema et al.'s (1998) data show that scour depth, relative to pier width, may increase with Euler number, V2/gb. They argued that the parameter V2/gb is useful for describing energy gradients for flow around a pier. It can be considered to express the ratio of stagnation head V2/2g to pier width. Flow-field similitude requires preservation of flow patterns such that pressure heads along flow paths scale directly with the geometric scale relating a model pier in the laboratory to a pier in the field. For the same stagnation head V2/2g, steeper gradients occur at narrower piers. A narrower pier will induce a smaller value of ds/b than a wider pier in the same flow field. More recently, Ettema et al. (2006) noted that a practical implication of the inability to concurrently scale pier size, flow depth and sediment size in flume experiments, is inadequate similitude of large-scale turbulence generated by flow around piers. They pointed out that the parameters V2/gb and Vb/ν can be interpreted as expressing similitude in the energy and frequency of eddies shed from the pier. The parameter V2/gb is in effect a normalised expression of vorticity of wake vortices. Because vorticity is roughly proportional to approach velocity V, the implication is that narrower piers in the same approach flow produce stronger eddies. Given the range of length scales commonly used in scour experiments, Reynolds number (Re) in terms of viscous effects is unlikely to have direct bearing on scour depth. However, Re also influences the frequency of shedding of vortices, n. For typical values of pier size and flow velocity used in flume experiments and also in the field, Re is in the range 103 to 105, for which the Strouhal number (nb/V) is about 0.2 for circular cylinders. Thus for piers in the same approach flow V, the frequency of shedding of vortices is inversely proportional to the pier size, i.e. narrower piers in the same flow generate eddies at a greater rate. The effects on local scour depth of inadequate scaling of strength and frequency of vortex shedding are shown schematically in Figure 3. The diagrams are consistent with the limited data presented by Ettema et al. (2006).

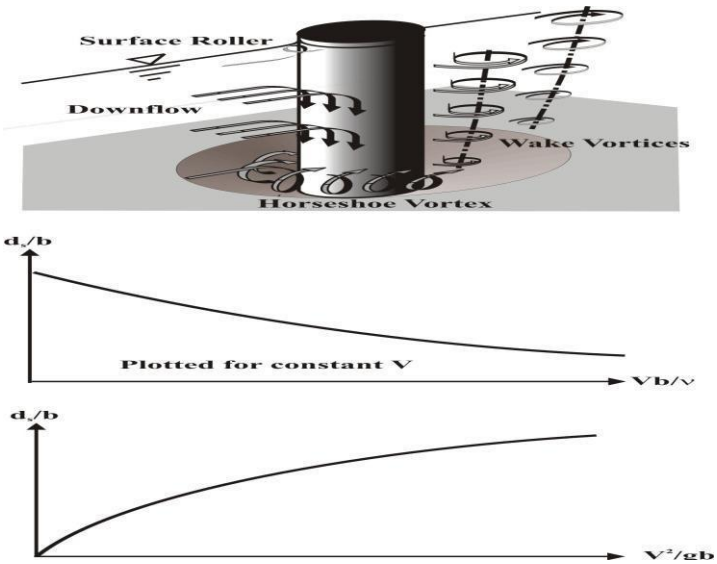

Fig.3 Local scour depth variation with large-scale turbulence

about 0.2 for circular cylinders. Thus for piers in the same approach flow V, the frequency of shedding of vortices is inversely proportional to the pier size, i.e. narrower piers in the same flow generate eddies at a greater rate. The effects on local scour depth of inadequate scaling of strength and frequency of vortex shedding are shown schematically in Figure 3. The diagrams are consistent with the limited data presented by Ettema et al. (2006).

4. Effect of Sediment Coarseness {local scour factors: sediment coarseness}, b/d50 :

The influence of sediment size on local scour depth is shown schematically in Figure 4. Data from small-scale laboratory experiments show that, for uniform sediments, local scour depths are unaffected by sediment coarseness unless the sediment is relatively large. Ettema (1980) explained that for smaller values of the sediment coarseness ratio (b/d50 < 50), individual grains are large relative to the groove excavated by the downflow and erosion is impeded because the porous bed dissipates some of the energy of the downflow. When b/d50 < 8, individual grains are so large relative to the pier that scour is mainly due to erosion at the sides of the pier and scour is further reduced. For much larger values of b/d50, representative of prototype sized piers founded in sandy materials, recent data by Sheppard et al. (2004) demonstrate significant scour depth reductions for increasing b/d50. The reductions for b/d50 > 50 are shown schematically in Figure 4. Sheppard et al. (2004) used three different diameter circular piers (0.114, 0.305 and 0.914 m), three different uniform cohesionless sediment diameters (0.22, 0.80 and 2.90 mm) and a range of water depths and flow velocities. The tests extended the range of ratios of b/d50 to 4155.

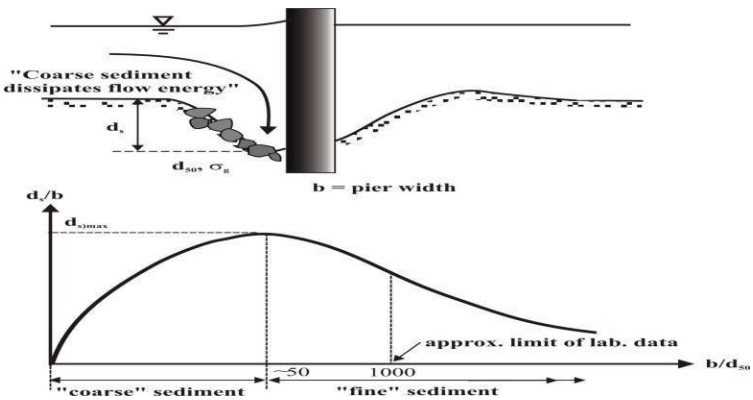

Fig.4 Local scour depth variation with sediment coarseness

5. Effect of Pier Shape :

Shape (multiplying) factors for simple piers, that is piers having constant section throughout their depth, have been proposed by several investigators, including Tison (1940), Laursen and Toch (1956), Chabert and Engeldinger (1956), Garde (1961), Larras (1963), Venkatadri et al. (1965), Yaroslavtiev (as given in Maza Alvarez, 1968), Dietz (1972), Neill (1973) and Richardson and Davis (1995). In practice, shape factors are only important if axial flow can be ensured. Even a small angle of attack will eliminate any benefit from a streamlined shape. Complex piers include piers with piled

foundations, caissons, and slab footings, and also tapered piers, as illustrated in Figure 8. For piers tapered on the upstream and downstream faces, the slope, in

elevation, of the leading edge of the pier affects the local scour depth. Downwards-tapering piers induce deeper scour than a circular pier of the same width, and vice-versa. Shape factors for local scour at tapered piers have been proposed by Neill (1973), Chiew (1984) and Breusers and Raudkivi (1991). For piers founded on a (wider than the pier) slab footing, caisson or pile cap, the footing, cap or caisson with the top below the general bed level can

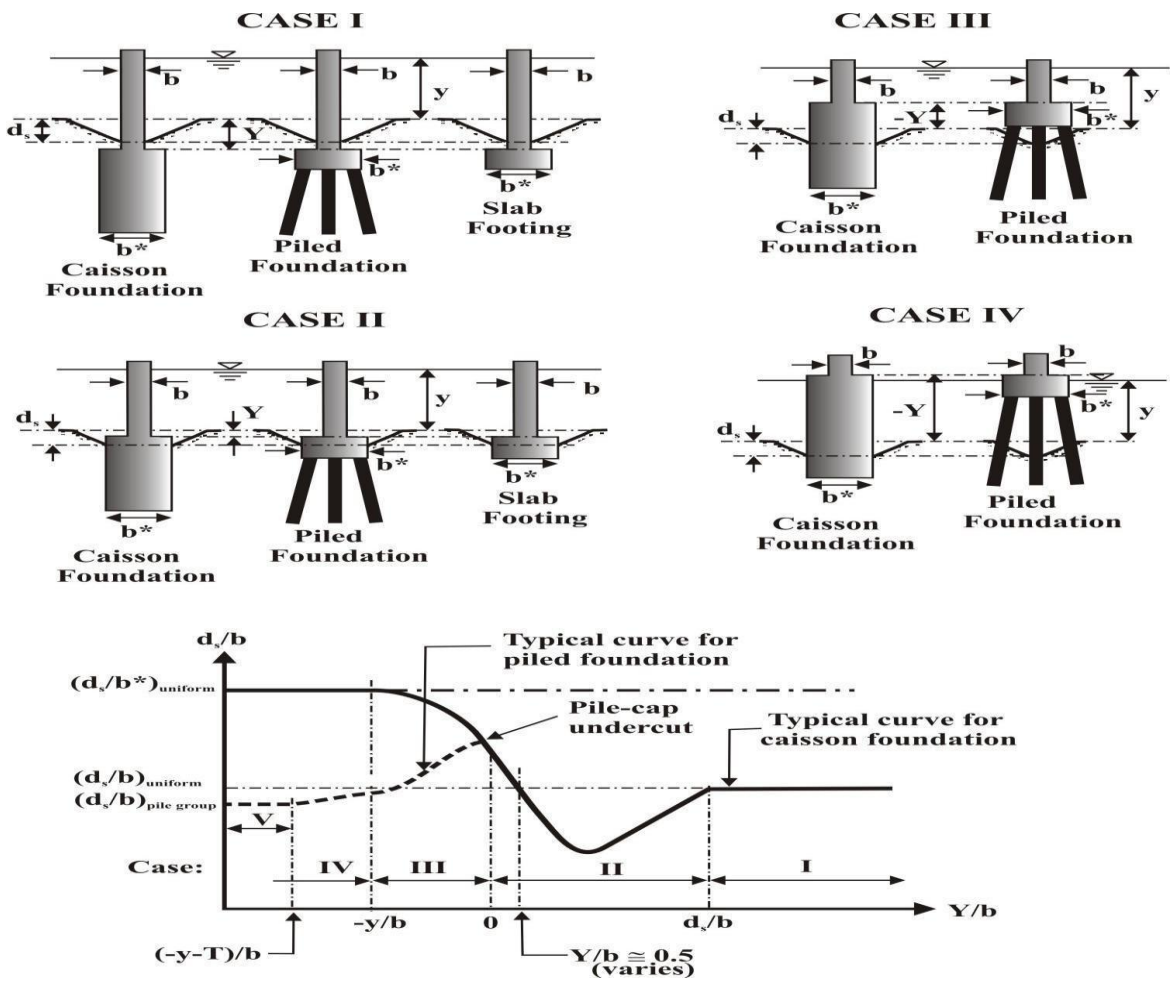

Fig.5 Local scour depth variation with complex pier shape

6. Methodology

The methodology outlines the systematic approach adopted to study how scour depth and local velocity fields influence the performance of bridge piers, particularly for offshore and large bridges.
This study combines a comprehensive literature review, data analysis, and model evaluation to understand the physical mechanisms, influencing factors, and mitigation strategies related to scour.

Research Framework :

The research methodology is divided into five main phases

1. Problem Identification and Objective Formulation
2. Literature Collection and Review
3. Data Extraction and Parameter Analysis
4. Model Evaluation and Comparative Study
5. Interpretation, Recommendations, and Conclusions

1. Methodology Workflow for Scour Depth and Velocity Field Study :

Phase	Activities / Tasks	Expected Output / Deliverable
1. Problem Identification	- Define research scope - Identify knowledge gaps in scour and velocity interactions - Set objectives	Clear research problem statement and objectives
2. Literature Collection & Review	- Gather journal papers, books, and guidelines (HEC-18, Melville & Coleman, Sumer & Fredsøe) - Review scour mechanisms, pier geometry effects, and velocity field influence	Comprehensive literature review, identification of critical factors
3. Data Extraction & Parameter Analysis	- Collect quantitative data: flow velocity, pier diameter, sediment size, maximum scour depth - Analyze relationships between local velocity and scour depth	Tabulated data, trends, and preliminary correlations
4. Model Evaluation & Comparative Study	- Compare empirical, semi-empirical, and CFD/numerical models - Evaluate applicability for offshore and large bridges - Analyze vortex formation and velocity field around piers	Comparative analysis, model suitability assessment, visualization of flow patterns
5. Interpretation & Recommendations	- Synthesize findings - Evaluate impact of scour on pier performance - Suggest mitigation strategies and monitoring techniques	Recommendations for bridge design and scour management, identification of research gaps

2. Different models were compared to assess their suitability for offshore and large bridges:

Model/Method	Type	Key Features	Applicability
CSU / HEC-18	Empirical	Riverine scour estimation	Limited to simple piers
Melville & Chiew (1999)	Semi-empirical	Accounts for pier shape and flow depth	Suitable for moderate flow

Sumer & Fredsøe (2002)	Mechanistic	Includes wave–current effects	Offshore applications

Model/Method	Type	Key Features	Applicability
CFD (ANSYS Fluent / OpenFOAM)	Numerical	Flow velocity, vortex simulation	High accuracy, complex setup

Tools and Software Used :

Tool/Software	Purpose
MS Excel / MATLAB	Data analysis and graph plotting
ANSYS Fluent / OpenFOAM	CFD-based flow field and scour simulation
AutoCAD / SketchUp	Diagram and schematic preparation

Result And Conclusion :

The relation between the depth of local scour at a bridge pier and its dependent parameters is discussed. The dependent parameters described are flow intensity (V/Vc), flow shallowness (y/b), Euler number (V2/gb), Reynolds number (Vb/ν), sediment coarseness (b/d50), sediment nonuniformity (σg), time (t), pier shape (for simple and complex piers, and for floating debris effects), pier alignment, and superstructure submergence. Emphasis is given to the underlying physics of the process of local scour. Knowledge gaps are noted. Recent research findings are included. In particular, Ettema et al. (2006) noted that it is impossible to currently scale pier size, flow depth and sediment size in flume experiments. A practical implication of this is inadequate similitude of large-scale turbulence generated by flow around piers. Also, recent data by Sheppard et al. (2004) demonstrate significant scour depth reductions for increasing b/d50 when b/d50 > 50. Thus, local scour depths at field scale may be significantly reduced from those observed in the laboratory.

Measured vs Predicted Scour Depths

Pier Diameter (D, m)	Flow Velocity (U, m/s)	Critical Velocity (Uc, m/s)	Velocity Ratio (U/Uc)	Measured Scour Depth (ds, m)	Predicted ds (HEC-18, m)	Predicted ds (CFD, m)
1.5	0.9	0.7	1.29	0.25	0.28	0.26
2	1.2	0.8	1.50	0.45	0.42	0.46
3	1.5	1.0	1.50	0.65	0.60	0.68

CFD predictions closely match measured scour depths, especially in offshore flow conditions.

Comparative Model Performance

Model	Accuracy	Strengths	Limitations
HEC-18 (Empirical)	Moderate	Easy to use; widely accepted for rivers	Limited for offshore/wave environments
Melville & Chiew	Moderate-High	Includes pier geometry effects treatment	Simplified turbulence
Sumer & Fredsøe	High	Accounts for wave-current and vortex effects	Requires detailed input data

Model	Accuracy	Strengths	Limitations
CFD (ANSYS/OpenFOAM)	Very High	Captures velocity field and turbulence accurately	Computationally intensive

Summary of Conclusions

Aspect	Key Findings / Observations	Implications / Recommendations
Velocity Effects	- Local flow velocity and velocity ratio (U/Uc) dominate scour depth. - High velocity zones correspond to maximum erosion.	- Accurate measurement of flow velocity is essential for pier design. - Consider extreme flow scenarios in offshore conditions.
Pier Geometry	- Circular vs rectangular piers influence vortex formation. - Larger piers induce deeper scour due to stronger horseshoe vortices.	- Select pier shape and diameter considering site-specific flow conditions. - Optimize geometry to reduce scour potential.
Sediment Characteristics	- Fine sediments erode faster; coarse sediments resist scour. - Offshore sediment beds often heterogeneous, affecting scour patterns.	- Incorporate sediment type in design and mitigation planning. - Conduct site-specific sediment surveys before construction.
Model Performance	- Empirical models (HEC-18) are moderately accurate for riverine conditions. - Semi-empirical models improve predictions with geometry considerations. - CFD provides highest accuracy for offshore piers, capturing vortices and turbulence.	- Use empirical/semi-empirical models for preliminary design. - Apply CFD for offshore or complex flow conditions.
Scour Mitigation	- Riprap, collars, geotextiles reduce local shear stress and limit erosion. - Placement effectiveness depends on flow and sediment characteristics.	- Design protective measures based on detailed flow and sediment analysis. - Combine mitigation with regular monitoring.
Monitoring & Safety	- Real-time monitoring improves detection of scour progression. - Early intervention prevents structural failure.	- Implement sensors or sonar mapping to track bed erosion. - Integrate monitoring data with design adjustments.
Future Research	- Hybrid models combining empirical, semi-empirical, and numerical approaches. - Study multi-pier interactions and wave-current effects. - Explore eco-friendly or adaptive scour protection.	- Enhance prediction accuracy while reducing computational costs. - Develop sustainable and site-specific mitigation strategies.

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